

RESEARCH INTERESTS

Advancing machine listening: understanding acoustic scenes and events and the emotions they evoke. My research topics include environmental sound event detection, real-life audio-visual scene recognition, audio-visual singing voice detection, and sound-related affective computing.

EDUCATION

Honorary Research Assistant Nov. 2022 — Mar. 2023

The Bartlett School of Environment, Energy and Resources, University College London (UCL), UK

Supervisor: Full Prof. Jian Kang

Ph.D. student in Computer Science Engineering Oct. 2020 — present

WAVES Research Group, Faculty of Engineering and Architecture, Ghent University, Belgium

Supervisor: Full Prof. Dick Botteldooren

Master of Electronics Science and Technology Sept. 2016 — Jun. 2020

Beijing University of Posts and Telecommunications (BUPT), China

Supervisor: Prof. Shengchen Li, Full Prof. Ming Liu

AWARD

- Chinese Government Award for Outstanding Self-financed Students Abroad July. 2024
(Recipients are chosen worldwide among all Chinese overseas graduate students for their outstanding accomplishments in academia; only up to 600 are awarded each year.)
- China National Scholarship Oct. 2014
- BUPT Honorary Scholarship Feb. 2019
- BUPT Scholarship for Postgraduate Studies Oct. 2016; Oct. 2017; Oct. 2018

PROFESSIONAL EXPERIENCE

Research intern at Speech Research Group, Supervisor: Prof. Frank K. Soong Jul. 2018 – Jun. 2019
Microsoft Research Asia (MSRA) Beijing, China

- Microsoft Xiaolce is an AI system that loves to sing. To better interact with users, Xiaolce needs to evaluate people's songs and understand real-life acoustic scenes. My research aims to detect the endpoints of singing voice in polyphonic songs.
- Outcome: one patent copyrighted by Microsoft (OBTAINING A SINGING VOICE DETECTION MODEL, patent ID: CN112309428A), and one paper accepted by INTERSPEECH 2020.

Research intern at Music Research Group, Supervisor: Dr. Bilei Zhu Nov. 2019 – Oct. 2020
ByteDance AI Lab Shanghai, China

- In live musical streams, an anchor is a person who speaks to the audience or sings in front of the camera while playing other music. This internship explores the feasibility of audio-visual singing voice detection in live video streams.
- Outcome: one paper accepted by ICASSP 2021, and another paper accepted by INTERSPEECH 2021.

PUBLICATIONS

[GOOGLE SCHOLAR](#)

Journal

- IEEE TASLP:** Y. Hou, B Kang, A Mitchell, W Wang, J Kang, D Botteldooren, "Cooperative Scene-Event Modelling for Acoustic Scene Classification".
- IEEE SPL:** Y. Hou, S Song, C Yu, W Wang, D Botteldooren, "Audio Event-Relational Graph Representation Learning for Acoustic Scene Classification".

3. **JASA:** Y. Hou, Q Ren, H Zhang, A Mitchell, F Aletta, J Kang, D Botteldooren, “AI-based soundscape analysis: Jointly identifying sound sources and predicting annoyance”.
4. **IEEE Robotics and Automation Letters:** Q Ren, Y. Hou, D Botteldooren, T Belpaeme, “No More Mumbles: Enhancing Robot Intelligibility Through Speech Adaptation”.
5. **JASA:** A. Mitchell, E. Brown, R. Deo, Y. Hou, “Deep learning techniques for noise annoyance detection: Results from an intensive workshop at the Alan Turing Institute”.
6. **Sensors:** Q Ren, Y. Hou, D Botteldooren, T Belpaeme, “Behavioural Models of Risk-Taking in Human–Robot Tactile Interactions”.
7. **Applied Acoustics:** A. Talebzadeh, D. Botteldooren, T. Van Renterghem, P. Thomas, D. Van de Velde, P. De Vriendt, Y. Hou, P. Devos, “Sound augmentation for people with dementia: Soundscape evaluation based on sound labelling”.

Selected Conference

1. **IEEE ICASSP 2024:** Y. Hou, Q Ren, S Song, Y Song, W Wang, & D. Botteldooren, “Multi-level graph learning for audio event classification and human-perceived annoyance rating prediction”.
2. **AAAI 2024:** Q Lin, C Luo, Z Niu, X He, W Xie, Y. Hou, L Shen, S Song, “Boosting Adversarial Transferability across Model Genus by Deformation-Constrained Warping”.
3. **IEEE ICASSP 2023:** Y. Hou, Y. Wang, W. Wang, D. Botteldooren, “GCT: Gated Contextual Transformer for Sequential Audio Tagging”.
4. **INTERSPEECH 2023:** Y. Hou, S Song, C Luo, A Mitchell, Q Ren, W Xie, J Kang, W Wang, D. Botteldooren, “Joint Prediction of Audio Event and Annoyance Rating in an Urban Soundscape by Hierarchical Graph Representation Learning”.
5. **ACM/IEEE HRI 2023:** Q Ren, Y. Hou, T. Belpaeme, “Low-latency Classification of Social Haptic Gestures Using Transformers”.
6. **ICONIP 2023:** Z Liu, Y. Hou, H Tang, Á López-Chilet, S Michiels, D Botteldooren, “CLF-AIAD: A Contrastive Learning Framework for Acoustic Industrial Anomaly Detection”.
7. **IEEE MMSP 2022:** Y. Hou, B. Kang & D. Botteldooren, “Audio-visual scene classification via contrastive event-object alignment and semantic-based fusion”.
8. **INTERSPEECH 2022:** Y. Hou, Z. Liu, B. Kang, Y. Wang & D. Botteldooren, “CT-SAT: Contextual Transformer for Sequential Audio Tagging”.
9. **INTERSPEECH 2022:** Y. Hou, & D. Botteldooren, “Event-related data conditioning for acoustic event classification”.
10. **IJCNN 2022:** Y. Hou, B. Kang, W. Hauwermeiren, & D. Botteldooren, “Relation-guided acoustic scene classification aided with event embeddings”.
11. **IEEE ICASSP 2021:** Y. Hou, Y. Deng, B. Zhu, Z. Ma, & D. Botteldooren, “Rule-embedded network for audio-visual voice activity detection in live musical video streams”.
12. **INTERSPEECH 2021:** Y. Hou, Z. Yu, X. Liang, X. Du, B. Zhu, Z. Ma, & D. Botteldooren, “Attention-based cross-modal fusion for audio-visual voice activity detection in musical video streams”.

REVIEWER

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- IEEE/ACM Transactions on Audio Speech and Language Processing (TASLP)
 - IEEE Transactions on Systems, Man, and Cybernetics: Systems (SMC: Systems)
 - Journal of the Acoustical Society of America (JASA)
 - IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)
 - Conference of the International Speech Communication Association (INTERSPEECH)
 - IEEE International Workshop on Machine Learning for Signal Processing (MLSP)
 - IEEE International Joint Conference on Neural Networks (IJCNN)
 - IEEE Conference on Artificial Intelligence (IEEE CAI)